

driver. This 8-channel source driver is useful for interfacing between low-level logic and high-current loads. Its maximum output current is 500mA.

UM66 (IC4) is a melody-generator IC. 7809 (IC5) is a 9V voltage regulator, which converts 12V to regulated 9V output.

LDR1 along with potmeter VR1 and resistor R1 form the voltage divider at pin 2 of IC1. Trigger voltage is fed to IC1 through this voltage divider. When light falls on LDR1, its resistance and the potential difference between pin 2 of IC1 and ground decreases. In this condition, output pin 3 of IC1 remains high, which causes LED1 to glow. Sensitivity of LDR1 can be adjusted using VR1.

When input pin 1 of IC2 becomes high, its output pin 18 becomes high. This means, NE555 (IC3) pins 4 and 8 also become high. This leads pin 2 of IC3 to get grounded through C1 and its pin 3 to go high.

PARTS LIST

Semiconductors:

IC1, IC3	- NE555 timer
IC2	- UDN2981
IC4	- UM66 melody generator
IC5	- 7809, 9V voltage generator
T1	- BC547 npn transistor
LED1, LED2	- 5mm LED
ZD1	- 3.3V Zener diode
LDR1	- 5mm LDR

Resistors (all 1/4-watt, $\pm 5\%$ carbon), unless stated otherwise:

R1	- 2.2-kilo-ohm
R2, R3, R6, R9,	
R11	- 1-kilo-ohm
R4, R5, R7, R8	- 10-kilo-ohm
R10	- 33-ohm
R12	- 33-ohm, 0.5W
R13	- 56-kilo-ohm
R14	- 100-kilo-ohm
R15	- 150-kilo-ohm
R16	- 220-kilo-ohm
R17	- 330-kilo-ohm
R18	- 560-kilo-ohm
R19	- 1-mega-ohm
R20	- 1.5-mega-ohm
VR1, VR2	- 100-kilo-ohm potmeter

Capacitors:

C1	- 10 μ F, 25V electrolytic
C2, C6, C8	- 10nF ceramic disk
C3	- 47 μ F, 25V electrolytic
C4	- 100 μ F, 25V electrolytic
C5	- 470 μ F, 25V electrolytic
C7	- 10 μ F, 16V electrolytic

Miscellaneous:

BATT.1, BATT.2	- 12V rechargeable battery
RL1	- 5V, 1CO relay
LS1	- 8-ohm, 0.5W speaker
M1	- 12V DC motor/watering pump
DIP1	- 8-way DIP switch
	- 2-pin terminal connector (2 Nos)

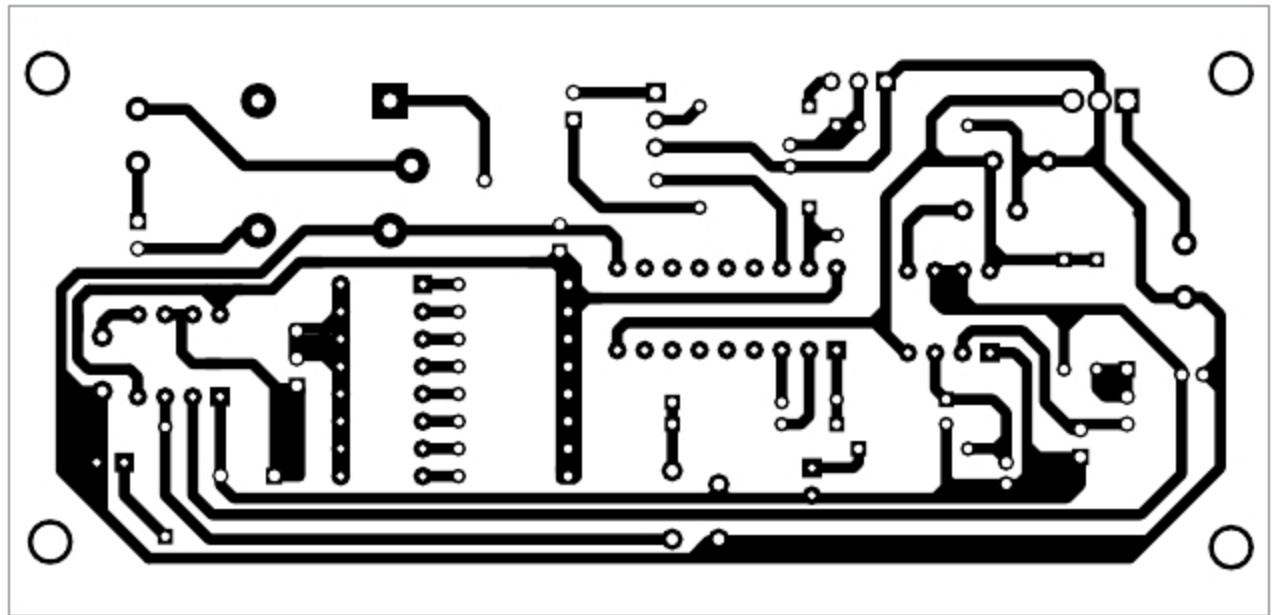


Fig. 3: Actual-size PCB layout of automatic plants watering system with melody

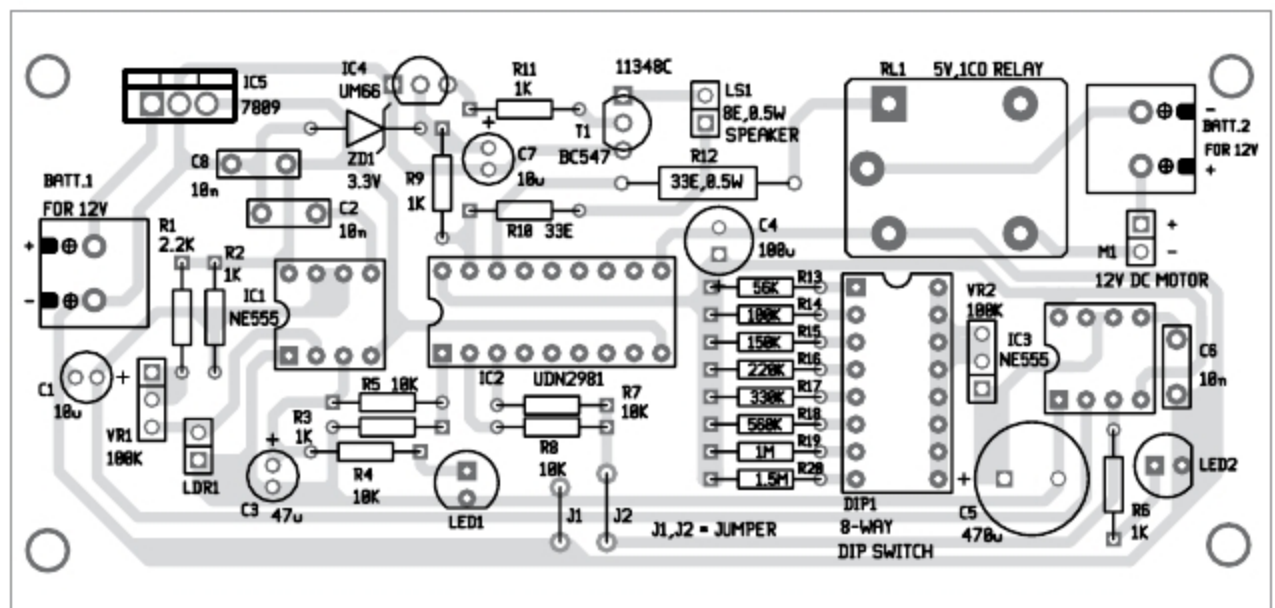


Fig. 4: Components layout for the PCB

This causes LED2 to glow for a certain period of time, which is determined by the values of C5 and R13 through R20. Actual on-time output of IC3 is shown in the table below, assuming that value of VR2 is zero.

When you add the value of VR2 to R13 through R20, this on-time may increase accordingly. When C5 charges to 2/3rd of supply voltage,

output pin 3 of IC3 goes low. Motor M1 for watering the plants and melody sound are off. DIP switch DIP1 is used for different time settings, as shown in the table.

When output pin 3 of IC3 is high, input pins 2 and 3 of IC2 become high through R7 and R8. This causes IC2 output pins 17 and 16 to become high. When output pin 16 of IC2 becomes high, current flows through R12 and relay RL1 is energised. Hence, a 12V DC pump motor fitted with a suitable pipe starts working. The amount of water for watering the plants depends on the type of DC pump motor used.

UM66's maximum working voltage is 3.3V. A 3.3V Zener diode is used to provide the maximum required voltage. UM66 gets input from UDN2981, and its output is fed to the base of transistor T1 through R11.

Resistor R10 is used to set the threshold operation of the circuit. When UM66 is enabled, it produces a

VALUES OF RESISTORS R13 TO R20 AND TIME

Resistor	Value	Time in seconds
R13	56k	27.9
R14	100k	57.7
R15	150k	77.5
R16	220k	113.7
R17	330k	170.6
R18	560k	289.5
R19	1M	512.0
R20	1.5M	775.5